

UNIT V

5633032

19. Explain Hash Indexing with an example. (11)

Or

20. Explain how the external sorting is performed with tapes using suitable example. (11)

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Third Semester

Computer Science and Engineering

DATA STRUCTURES

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define ADT.
2. List out and define the performance measures of an algorithm.
3. List the stack operations.
4. What are the advantages of Linked List over arrays?
5. Define a full binary tree.
6. Define tree traversal.

7. Define minimum spanning tree.
8. List the properties of set equivalence relation.
9. What is Rehashing?
10. Define External sorting.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the best case and worst case analysis of the binary search. (11)

Or.

12. Explain the quick sort algorithm and give the time complexity analysis in worst case. (11)

UNIT II

13. Explain the array and linked list implementation of Queue. (11)

Or

14. Explain how stack is applied for evaluating an arithmetic expression. (11)

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UNIT III

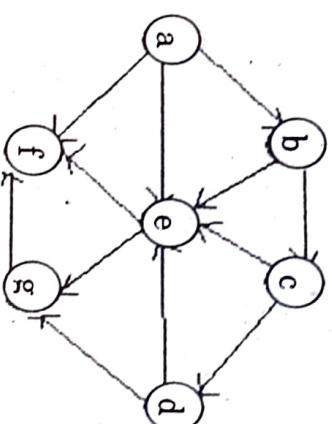
15. Write an algorithm to insert, delete, find minimum and maximum element from a binary search tree.

Or

16. Explain the AVL rotations with suitable example.

UNIT IV

17. What is Topological sort? Write down the pseudo code to perform topological sort, and apply the same to the following graph.



Or

18. Explain the various operations on Sets with suitable example.

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